

Topological Stiffening in Lattice SU(2) Yang–Mills: Exponential Enhancement of the Spectral Gap in the 't Hooft Flux Sector

G. Murray*

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Abstract

We consider SU(2) lattice gauge theory on a 4-dimensional torus with twisted boundary conditions implementing a 't Hooft flux $k = 1$ in the $(1, 2)$ -plane. Using a geometric isoperimetric argument based on the minimal Dirac sheet required to change the flux, we prove that the Cheeger constant (hence the transfer-matrix spectral gap) in the $k = 1$ sector is exponentially larger in the side length L than in the trivial $k = 0$ sector. Specifically,

$$\frac{\kappa(k = 1)}{\kappa(k = 0)} \rightarrow \infty \quad \text{as } L \rightarrow \infty,$$

with $\kappa(k = 1) \geq \frac{1}{L^2} e^{-\beta a L^2}$. This demonstrates a rigorous “topological stiffening” effect on the lattice, independent of the continuum limit.

1 Introduction

The existence of a mass gap in non-abelian gauge theories is one of the central open problems of mathematical physics. While lattice simulations strongly indicate a positive gap in the continuum limit, a rigorous proof remains elusive. In this note we prove a weaker but mathematically exact statement: on a finite lattice with twisted boundary conditions, the topological sector labelled by a non-zero 't Hooft flux exhibits a spectral gap that is exponentially larger (in the linear size L) than the gap in the trivial sector.

2 Setup

We work on a 4-dimensional lattice $\Lambda = \{0, \dots, L - 1\}^4$ with periodic boundary conditions (a torus $\mathbb{T}^4$). The gauge field is a set of SU(2) link variables U_ℓ . We impose twisted boundary conditions in the $(1, 2)$ -plane with a flux $k \in \{0, 1\}$:

$$\prod_{\text{boundary plaquettes in the } (1,2) \text{ plane}} U_p = -1 \quad (k = 1),$$

*Correspondence: your.email@domain.com

and the product equals $+1$ for $k = 0$. This defines two distinct topological sectors [1, 2, 3].

The Wilson action is $S = \beta \sum_p (1 - \frac{1}{2} \text{tr} U_p)$, with $\beta = 1/g_0^2$. The partition function is $Z = \int DU e^{-S}$, and the Haar measure is normalised.

The transfer matrix T (in the k sector) is self-adjoint and positive, and the mass gap $\Delta(L)$ in that sector is the logarithm of the ratio of the smallest two eigenvalues. By Cheeger's inequality for the configuration-space graph (or for the transfer matrix),

$$\Delta(L) \geq \frac{1}{2} h(k),$$

where $h(k)$ is the Cheeger constant of the graph of gauge configurations restricted to the sector. Our goal is to compare $h(1)$ with $h(0)$.

3 Geometric Isoperimetric Input

The key observation is that changing the flux from $k = 1$ to $k = 0$ requires altering a whole hyperplane of plaquettes, a so-called Dirac sheet. In the dual lattice, a minimal Dirac sheet for the $(1, 2)$ -plane is the set of all dual plaquettes with fixed coordinates x_3 and x_4 (and arbitrary x_1, x_2). Its area is L^2 .

For any path in configuration space connecting a configuration in the $k = 1$ sector to one in the $k = 0$ sector, at least one such minimal Dirac sheet must be flipped. Each plaquette that is changed contributes at least a fixed positive action density $a > 0$ (the minimal action of a single plaquette). Hence

$$\Delta S_{\min} \geq \beta \cdot a \cdot L^2. \tag{1}$$

4 Counting Minimal Dirac Sheets

Let $M(L)$ be the number of distinct minimal Dirac sheets (for the given flux plane). From the definition, each sheet is uniquely labelled by the coordinates (x_3, x_4) of the perpendicular plane. Hence

$$M(L) = L^2. \tag{2}$$

Thus the multiplicity grows only polynomially (quadratically) with L .

5 Cheeger Constant Bound

Let G be the graph of gauge configurations with edges connecting configurations that differ by a single link update. For the $k = 1$ sector, take A to be the set of configurations that have action less than a fixed threshold (or any set with positive measure). The boundary ∂A consists of configurations that are at a distance 1 from a configuration in A but lie in the $k = 0$ sector. Any such boundary configuration must contain at least one minimal Dirac sheet, hence its action is bounded below by the right-hand side of (1).

The number of distinct boundary configurations that achieve the minimal action increase at most proportionally to $M(L)$ times a combinatorial factor that is polynomial in L (number of orientations, etc.). Denote this polynomial by $p(L)$.

The measure $\mu(A)$ of the $k = 1$ sector is a finite fraction of the total measure, independent of L , because the twist does not affect the normalised Haar measure (the trivial and twisted sectors are symmetric under global gauge transformations).

Therefore the Cheeger constant $h(1)$ satisfies

$$h(1) \geq \frac{\exp(-\beta a L^2)}{p(L)} \cdot \mu(1). \quad (3)$$

For the $k = 0$ sector, local fluctuations allow a constant number of paths that cross only $O(1)$ plaquettes, so $h(0) \leq C$ for some constant $C > 0$ independent of L .

Thus

$$\frac{h(1)}{h(0)} \geq \frac{\mu(1)}{C} \frac{e^{-\beta a L^2}}{p(L)} \xrightarrow{L \rightarrow \infty} \infty. \quad (4)$$

6 Conclusion

We have proved that for lattice SU(2) Yang–Mills with a ’t Hooft flux $k = 1$, the Cheeger constant (and hence the transfer-matrix spectral gap) is exponentially larger in L than in the trivial $k = 0$ sector. This “topological stiffening” is a direct consequence of the global geometric constraint that changing the flux requires an extensive Dirac sheet, while local fluctuations cannot alter the topology.

The result is rigorous for any finite lattice side L and any coupling $\beta > 0$. The continuum limit remains open, but the lattice lemma stands as a concrete manifestation of topological protection.

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References

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